

RenderMan Rendering

1. Project Aim

The goal of this project is to use RenderMan and RIB files to create a simple but complete still life rendering scene. According to the coursework requirements, the scene does not need to be very complex, but it should show an understanding of modelling, materials, textures, lighting, render settings, and final image control. Therefore, I did not choose a very large or complicated scene. Instead, I focused on a small still life composition: a glass cup, red juice, ice cubes, coins, and a wooden table.

2. Scene Object Selection and Overall Composition

I chose the glass cup as the main object because glass material is quite challenging. It involves many parameters, such as reflection, refraction, transparency, IOR, and roughness. If I only rendered a normal opaque object, the material adjustment would be much easier. However, the glass cup can better show how the renderer handles light.

The juice is placed inside the cup, and its colour is set to a red tone. On one hand, this colour creates contrast with the transparent glass. On the other hand, it also makes the liquid more visible under the lighting. If the liquid is too transparent, the image will lose its visual focus. If it is too dark, it will look like a solid piece of plastic. Therefore, I kept adjusting the absorption, turbidity, and specular parameters of the juice, so that it could keep its colour while still allowing light to pass through.

The ice cubes were an important object for later improvement. At first, the ice cubes looked like normal transparent blocks, and they did not have enough internal ice crystals, cracks, or frosted feeling. Therefore, I later modified the OSL shader for the ice cubes, adding internal cloudy effects, fine cracks, and slight surface bump. In this way, the ice cubes are not just transparent objects, but have more natural details.

I also added coins and a wooden table to the scene. The coins help to make the image feel more like a real still life scene, and they also provide a contrast with the metal material. The wooden table is the supporting surface of the whole scene. I added a wood texture, normal map, and displacement map to it, so the tabletop would not look too flat.

3. Modelling and File Structure

The project is mainly organised through “RIB files”. The main file is `scene.rib`, and the other objects are placed in the `assets` folder, such as the glass cup, juice, ice cubes, coins, and table. The advantage of this structure is that it is clear and easy to manage. Each object can be modified separately, and it is also easier to find and fix problems.

```
RenderProject/
├─ scene.rib
├─ scene_exr_only.rib
├─ assets/
│  ├─ glass.rib
│  ├─ juice2.rib
│  ├─ ice.rib
│  ├─ coin.rib
│  └─ table.rib
├─ shaders/
│  ├─ realistic_juice.osl
│  ├─ realistic_wood_table.osl
│  ├─ ice_crystal.osl
│  └─ coin_metal.osl
├─ textures/
│  ├─ sky.tex
│  ├─ wood.tex
│  ├─ table_normal.tex
│  ├─ table_displacement.tex
│  └─ coin_normal.tex
└─ renders/
```

There was originally a `Sphere.rib` file in the project, but I did not continue to use it in the final scene. This is because I wanted the visual focus to stay on the glass cup and the ice cubes, and I did not want extra objects to distract the viewer.

4. Glass Cup Material Creation Process

The glass cup is one of the most important materials in the whole project. At the beginning, although the glass cup was transparent, the overall result looked a little flat and stiff. The highlight on the edge was also too strong. In particular, there was often a very obvious white highlight in the lower-right area.

At first, this problem seemed to be caused by the ice cubes. However, after observing its position, I found that the white light was mainly attached to the edge of the glass cup. Therefore, it was more likely to be the Fresnel highlight and refraction highlight from the glass itself.

This step was very important, because it made me realise that when dealing with transparent materials, I should not only look at the surface position. I also need to judge which object and which angle the light comes from. Glass naturally has strong reflection at the edge angle. If the HDR brightness, `reflectionGain`, and `roughness` are not controlled well, an overexposed white edge can easily appear.

Therefore, I made several adjustments to the glass material later. The first step was to reduce the overly strong `reflectionGain` and HDR exposure, so the white highlight would not be too harsh. The second step was to slightly increase the glass `roughness`, changing the highlight from a hard bright edge into a softer bright line. The third step was to increase part of the `refractionGain` again after reducing the white highlight, so the glass would not become too grey or dirty.

The final goal was not to remove the highlight completely, but to make it look like a real edge reflection on glass. This process can be summarised as: overly bright highlight → reduce reflection and environment light → glass becomes darker → increase transparency and brightness again → find a balance.

5. Juice Material Creation Process

At the beginning, the main problem with the juice material was that the colour was too dark. It looked more like a red solid object rather than a transparent liquid. To make it look more like real juice, I adjusted it in several ways.

First, I adjusted the colour. The juice should not only use a simple red diffuse colour, because liquid mainly shows its thickness through light transmission and absorption. Therefore, I kept the diffuseGain relatively low and relied more on specular, refraction, and absorption to create the liquid effect.

Second, I adjusted the absorptionDepth. This parameter affects how the colour becomes deeper when light passes through the liquid. If the absorption is too strong, the liquid becomes very dark. If the absorption is too weak, the liquid loses its sense of thickness. In the final version, I adjusted it into a relatively bright red colour, so the cup edge and the inside of the liquid can both show some layers. However, the final result still looks more like a milkshake or thick yoghurt than clear juice.

6. Ice Cube Material Creation Process

The ice cubes were one of the parts that I improved the most in the later stage. At the beginning, the ice cubes

looked more like normal glass blocks. Although they were transparent, they did not have enough ice characteristics.

Real ice cubes are usually not completely clear. They often have small cracks, bubbles, cloudy structures, and surface scratches inside. To achieve this effect, I did not mainly depend on external textures. Instead, I used an OSL procedural shader to generate the ice details.

Several procedural effects were added in the shader, including cloudy core for the internal cloudy feeling, fine cracks for small cracks, frost edge for the frosted edges, and surface bump for slight uneven details on the surface.

7. Wooden Table, Coins, and Texture Processing

The wooden table is used to give the whole scene a stable background. To avoid making the tabletop look too flat, I used a wood texture, a normal map, and a displacement map. The wood texture provides colour variation, the normal map provides surface direction changes, and the displacement map creates slight unevenness on the table surface.

Although the wood grain mainly comes from texture maps, it can create a more natural tabletop effect when combined with shader parameters. The table does not need to be the most eye-catching object, but it needs to support the main objects and should not look empty in close-up shots.

The coins use a metal material, with a normal map added to show surface details. Their purpose is not to become the centre of the image, but to provide material contrast. The glass and ice cubes are transparent materials, the juice is a semi-transparent liquid, and the coins are reflective metal objects. This makes the material types in the whole image richer.

8. Lighting, HDR, and Camera

For the lighting, I used an “HDR environment map” as the main light source. The advantage of HDR is that it can provide more natural environment reflections. This is especially important because glass, ice cubes, and liquid all depend strongly on reflections. If I only used normal point lights, the glass would lose many realistic reflection changes.

However, HDR also created a problem. If there are very bright areas in the environment map, the edges of the glass and the surface of the ice cubes can easily become overexposed. Therefore, I adjusted the HDR ‘exposure’ several times in the later stage. I first reduced it from a brighter level, and then balanced it again according to the transparency of the glass and juice.

For the camera, I added a slight depth of field effect. The main parameter is: ‘DepthOfField 6.5 50 33.0’

The first value mainly controls the strength of the blur. A smaller value creates a stronger background blur, while a larger value makes the image clearer. I did not use a very strong depth of field, because this project needs to show material details. If the depth of field is too strong, the ice cracks, glass edges, and juice layers will become blurred. Therefore, I finally chose a light depth of field effect, keeping the main objects clear while making the background slightly softer.

9. Final Image

The final image is a small still life scene. There is pink juice inside the glass cup, with an ice cube placed next to it.

The scene is lit by an HDR environment map. The main focus of the image is on the glass and liquid materials. The ice cubes provide extra details, while the wooden table and coins provide environment support and material contrast.

At the same time, the development process also shows how I moved from problems to solutions. For example, the strong white highlight in the lower-right area of the glass, the ice cube shader compiling problem, and the black EXR output problem were not solved by simply changing one parameter. I needed to judge the cause, test different methods, and then choose the most stable solution.

This project helped me understand the full workflow of RenderMan still life rendering more clearly. From the first reference analysis, to object organisation, shader writing, texture usage, lighting adjustment, depth of field setting, and final output troubleshooting, the whole process was not completed in one step. It was gradually built through many tests and modifications.